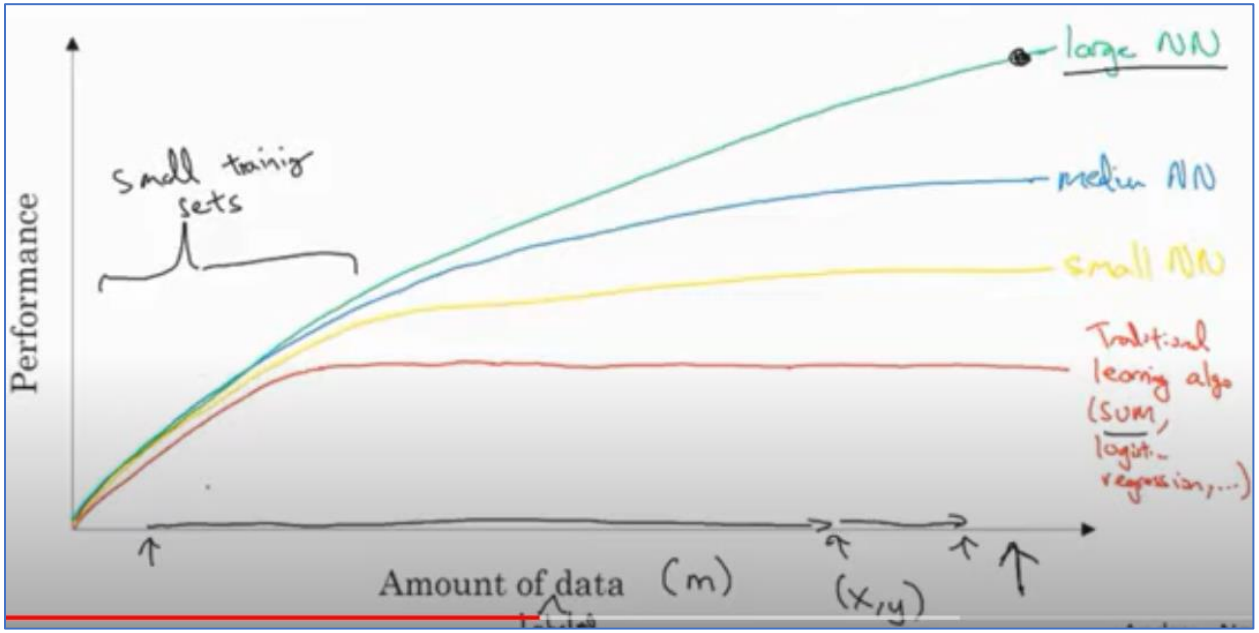
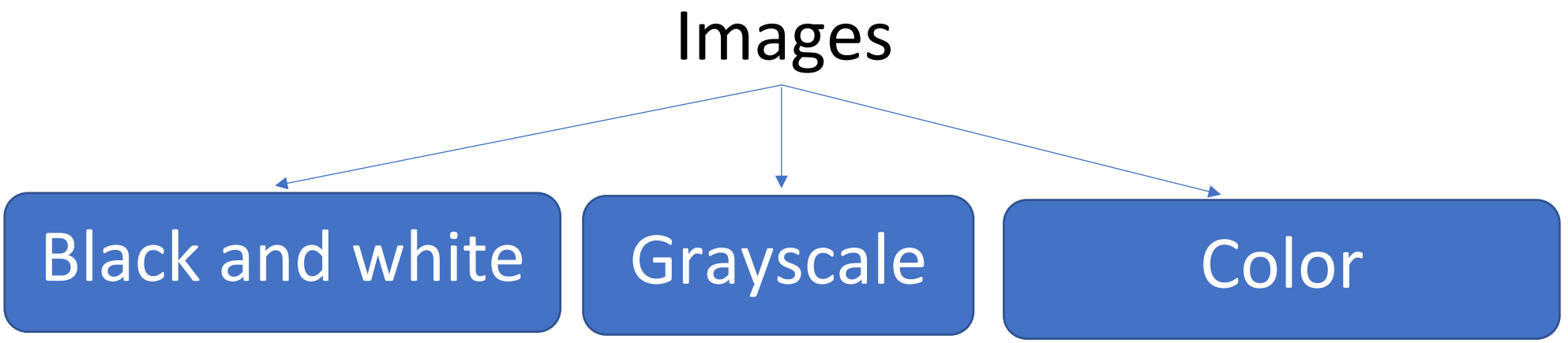
Two observations if you want to hit this very high level of performance:

- you need to be able to train a big enough neural network in order to take advantage of the huge amount of data.
- you need a lot of data

so we often say that scale has been driving deep learning progress and by scale we mean both the size of the neural network and the amount of data.



Classification of Images using NN









Examples

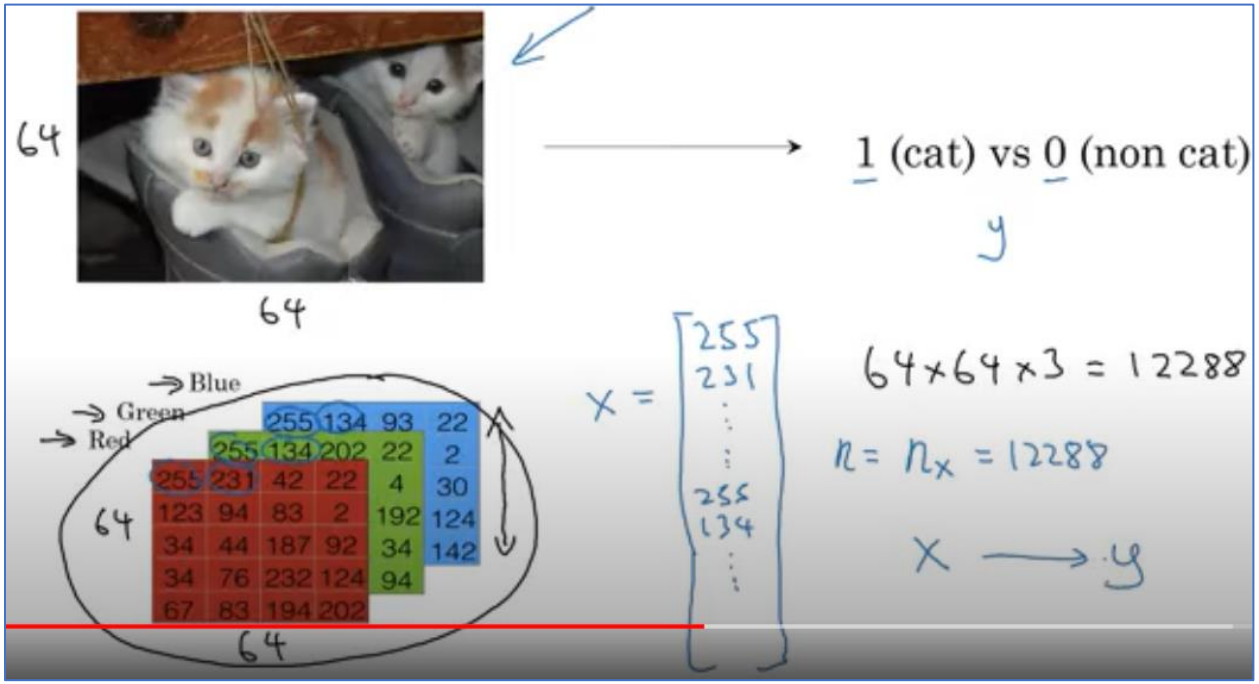
Black and white (Zero and one)

0	0	0	0	0	0	0
1	1	1	1	1	1	1
0	0	0	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	0	0	0

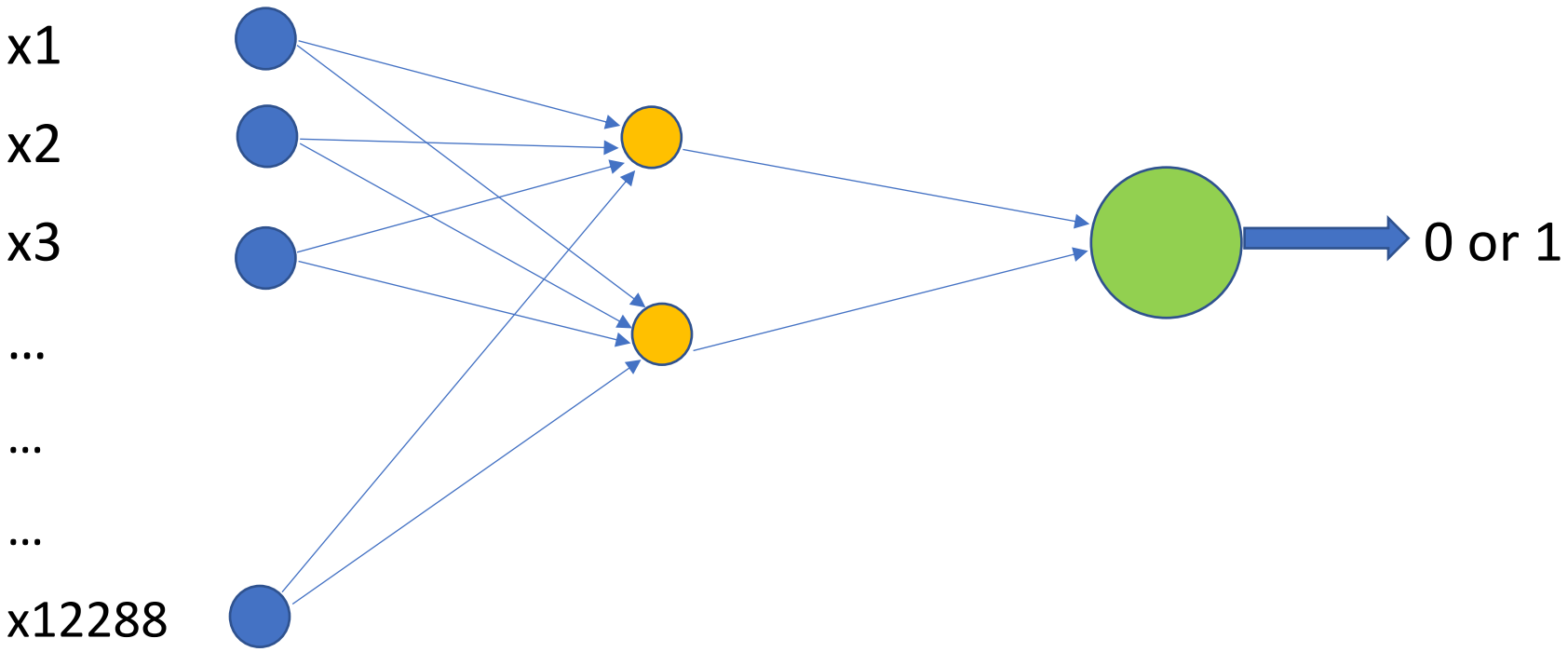
Grayscale ([0 ~ 255])

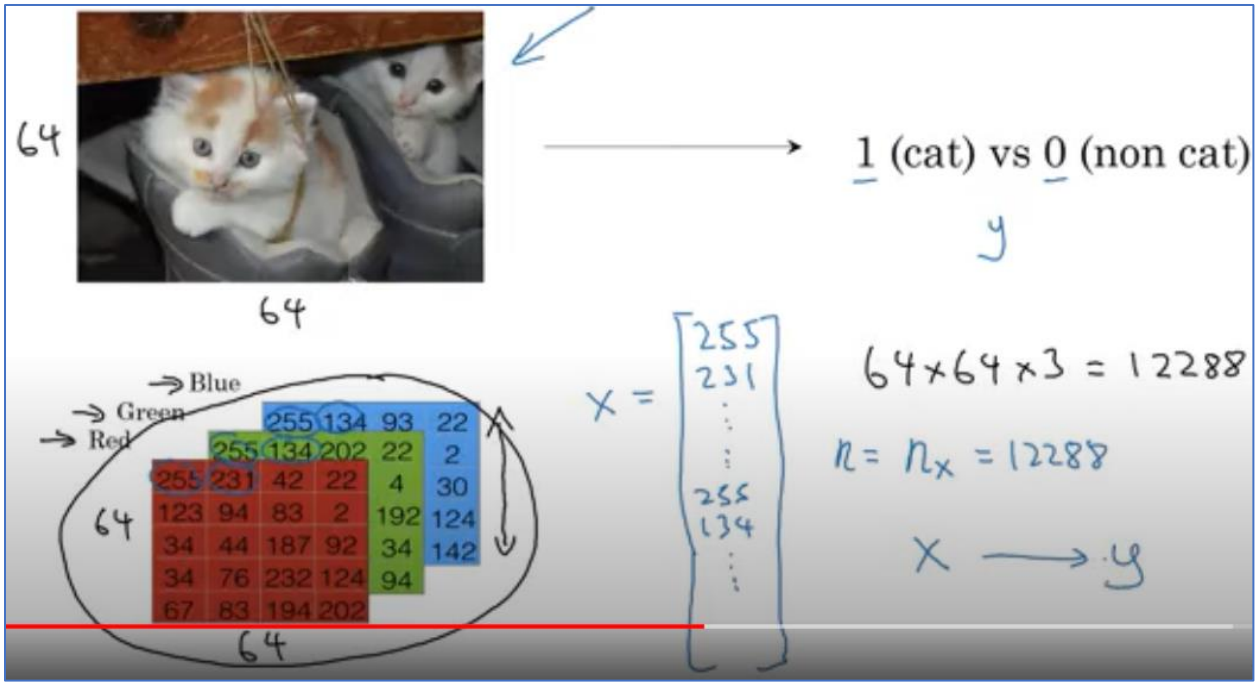
201	165	20	39	99	111	24	17	87	93
2	19	25	77	18	0	0	0	0	203
89	144	136	185	143	20	39	74	65	18
255	255	255	255	255	255	255	255	255	255
12	17	18	99	53	158	215	196	7	0
87	187	29	27	111	200	15	19	27	58
201	165	20	39	99	111	24	17	87	93

Color images (RGB)



of inputs = 12288


x1
x2
x3
...
...
...
x12288
0 or 1



- A single training example is represented by a pair (x,y) where x is an N dimensional feature vector and y is the label (zero or one).
- To put all the training examples into a more compact notation we are going to define a matrix (X).

(x, y) $\quad x \in \mathbb{R}^{n_x}$, $y \in \{0, 1\}$

m training examples: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(m)}, y^{(m)})\}$

$M = M_{train}$ $\qquad M_{test}$ = #test examples.

$$X = \begin{bmatrix} | & | & & | \\ x^{(1)} & x^{(2)} & \cdots & x^{(m)} \\ | & | & & | \end{bmatrix}$$

n_x

m

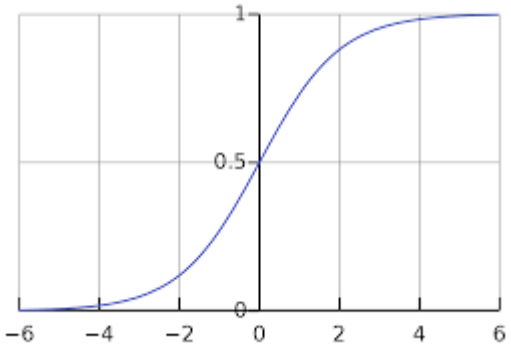
$X \in \mathbb{R}^{n_x \times m}$ $\qquad X.shape = (n_x, m)$

$$Y = \begin{bmatrix} y^{(1)} & y^{(2)} & \cdots & y^{(m)} \end{bmatrix}$$

$Y \in \mathbb{R}^{1 \times m}$

$Y.shape = (1, m)$

Sigmoid function



$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Hints:

- $e^{-z} = 1 / e^{z}$ (e: is mathematical constant approximately 2.71828)

- (1 / very large number) approaches zero